INCH-POUND

MIL-PRF-1/1654B(NAVY)
9 July 1998
SUPERSEDING
MIL-E-1/1654A(NAVY)
9 November 1970

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING-WAVE
TYPE 8364

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Forward-wave amplifier, frequency range 5.4 to 5.9 GHz, 200 mW power output, periodic-permanent-magnet focus, convection cooled, coaxial r-f connectors.

Outline:	See figure 1.
Mounting position:	Any.
Cooling:	Convection.
Connectors, r-f:	See figure 1.
Connectors, power:	See figure 1.
Focusing circuit:	PPM integral.
Weight:	3 pounds.

ABSOLUTE RATINGS: 1/ 2/

Parameter: / Unit:	Ef / V	If / A, surge	Ec / V	Ic / mA	Ew / V	Ik / mA
Maximum	11.5	4.0	30	1.0	1,350	15
Minimum	10.5	---	-250	---	---	---
Test condition: (1):	11.0	---	5/	---	5/	---
Test condition: (2):	0	---	0	---	0	---

Parameter: / Unit:	Pp / W	Pi(rf) / dBm	tk / sec	TA / °C	TE / °C 3/
Maximum	17	20	---	+75	+150
Minimum	---	---	60	-55	-55
Test condition: (1):	---	25/	---	25± 5	---
Test condition: (2):	---	0	---	---	---

GENERAL:

Qualification - Required.
Marking: 4/

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:13Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection.

Inspection	Method	Conditions	Symbol	Limits Min	Limits Max	Unit
Qualification inspection 26/						
Temperature	---	Test condition (1): F = 5.9 GHz 10/	---	---	---	---
Low temperature turn-on	---	Test condition (1): F = 5.9 GHz 11/	---	---	---	---
Humidity	---	Test condition (1): F = 5.9 GHz 12/	---	---	---	---
Temperature, non-operating	---	Test condition (2): 13/	---	---	---	---
Initial life test	---	Test condition (1): F = 5.9 GHz 19/ 23/ 24/ 28/	---	---	---	---
Conformance inspection part 1 27/						
Heater current	1301	Ef = 11.0 V	If	0.73	1.05	A
Large-signal power output	4250	Test condition (1); F = 5.4, 5.5, 5.6, 5.7, 5.8, 5.9 GHz 6/ 7/ 8/ 21/	Pi(rf) Pso	-16 23	--- ---	dBm dBm
		F = 5.4 to 5.9 GHz F = 5.4 to 5.65 GHz F = 5.65 to 5.9 GHz 9/	ΔPo ΔPo ΔPo	--- --- ---	2.0 1.5 1.5	dB dB dB
			Ew Ec	1,100 -9	1,250 0	V V
Noise figure	4260	Test condition (1): F = 5.4, 5.65, 5.9 GHz	NF	---	35	dB
Control-electrode current	1256	Test condition (1): Pi(rf) = 0	Ic	0	0.5	mA
Cathode current	1256	Test condition (1): Pi(rf) = 0; tk = 60 sec	Ik	6	10	mA
Conformance inspection part 2 14/ 15/						
Spurious power output (1)	4261	Test condition (1); Pi(rf) = 0; F = 4.9 to 6.4 GHz 16/	spo	---	-10	dBm
Spurious power output (2)	4261	Test condition (1); Pi(rf) = 0; F = 4.9 to 6.4 GHz 16/	spo	---	-10	dBm

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:13Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

Inspection	Method	Conditions	Symbol	Limits		Unit
				Min	Max	
Conformance inspection part 2 - Continued 14/ 15/						
Power output variation	4250	Test condition (1); F = 5.4, 5.9 GHz; Ef = 10.5 to 11.5 V 17/	ΔPo	---	1.0	dB
		ΔEw = ± 1 percent 18/	ΔPo	---	1.5	dB
Direct interelectrode capacitance	4266	10/	Cc-all	---	50	pF
Vibration		Test condition (1); F = 5.65 GHz 21/	---	---	---	---
Shock	1042	Test condition (2); 22/	---	---	---	---
Conformance inspection part 3 20/						
Life test	---	Test condition (1); Group S F = 5.9 GHz 23/ 24/	t	5,000	---	hrs
Life test end point	---	Test condition (1); F = 5.9 GHz Pi(rf) = 0 dBm, max	Po	20	---	dBm

1/ Typical operating conditions for this device are described by test conditions (1). Because of the number of interrelated conditions possible with traveling-wave tubes, ratings shall refer to each individual rating with the tube otherwise operating in the above test condition. Note that ratings are not of themselves test conditions or operating conditions.

2/ Cathode and one leg of heater shall be connected internally. Shell, helix, and collector shall be at ground potential. All voltages are referred to cathode.

3/ TE is the temperature measured at point “A” on figure 1.

4/ Each tube shall be marked with the following information:

a. Serial number.

b. Cathode voltage for large-signal operation, Ew.

c. Grid voltage for large-signal operation, Ec.

d. Cathode current for large-signal operation.

e. Input and output r-f connector identification.

5/ Ec and Ew under test condition 1 are defined as the voltages at which the tube produces large-signal power output within the specified limits. These voltages shall be identical with the values specified in 4/.

6/ A filter shall be inserted between the tube and the power-measuring device to attenuate harmonic-frequency components by not less than 30 dB relative to the fundamental frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:13Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

7/ At each of the frequencies specified, with Pi(rf) adjusted to obtain saturated power output, Pso, the values of Pi(rf) and Pso shall be within the limits specified. Vary Pi(rf) above and below the saturation value to obtain the Pso-3 dB points. The values of Pi(rf) at these points shall be within the specified limits and shall differ by at least 9 dB.

8/ Adjust Pi(rf) to obtain saturated power output. Using this value of Pi(rf), measure Po continuously across the band 5.4 to 5.9 GHz. The total variation in power output, ΔPo, across the band shall be not greater than the limit specified.

9/ Cc-all is the capacitance between grid and all other elements, including the shell. The grid lead shall be at maximum separation from the shell and all others leads.

10/ The temperature test shall be conducted in accordance with the requirements of MIL-E-16400(NAVY), for class 4 equipment. The tube shall operate within the limits specified under conformance inspection, part 1, large-signal power output.

11/ With the tube in the environment chamber, reduce the chamber temperature to 0°C ± 2°C, and maintain the chamber temperature for at least 24 hours with the tube in the non-operating condition. At the end of the 24-hour period, turn on the tube. The tube shall operate within the limits specified under conformance inspection, part 1, large-signal power output, within 25 minutes of application of heater voltage.

12/ The humidity test shall be conducted in accordance with the requirements of MIL-E-16400(NAVY), for class 4 equipment. Reference measurements and all subsequent measurements shall consist of the test for large-signal power output under conformance inspection, part 1.

13/ The non-operating temperature test shall be conducted in accordance with the requirements of MIL-E-16400(NAVY). Upon completion of the test, the tube shall operate within the limits specified under conformance inspection, part 1, large signal power output.

14/ The conformance tests, part 2, shall be performed in accordance with MIL-STD-105, inspection level 2, with an acceptance level of 4 percent. In the event of lot rejection, tightened inspection shall be invoked. Normal inspection shall be resumed when two consecutive lots have conformed with the tests of conformance inspection, part 2. For purposes of inspection, the lot size shall be one month's production.

15/ Tubes shall be selected at random from the production lot, and shall previously have passed the test of conformance inspection, part 1.

16/ For spurious power output (1), the output termination shall have a nominal value of 50 ohms, and the input shall be terminated in a variable-phase short circuit. For spurious power output (2), the terminations shall be transposed. A preselector having a passband of not more than 30 MHz, and a tuning range of 4.9 to 6.4 GHz shall be used between the tube and the power-measuring device. The spurious power output, spo, shall be not greater than the specified value in any portion of the above frequency range at all phases of the short circuit.

17/ At each of the specified frequencies, adjust the r-f input power Pi(rf) to obtain saturation. Reduce the heater voltage to the lower of the specified values and read the power output, Po. Increase the heater voltage to the higher of the specified values and read the power output. Allow the power to stabilize at each reading. The total change in output shall be within the specified limit.

18/ At each of the specified frequencies adjust Pi(rf) to obtain saturation. Reduce Pi(rf) by 4 dB. Vary Ew from Ew - 1 percent to Ew +1 percent and note the power output, Po, at each value. Increase Pi(rf) to the saturation value plus 4 dB and repeat the power measurements. The change in power, ΔPo, from the reference value, Po, at Ew, in each case shall be within the limit specified.

19/ Tubes for these tests shall be selected at random from the production lot, and shall previously have passed the tests of conformation inspection, part 1.

20/ The tube shall meet the power output conditions at all frequencies within the band when operating into a load of l.5:1 VSWR, minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:13Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

21/ The tube shall be rigidly mounted to the vibration table and shall be vibrated in accordance with the requirements of MIL-STD-167 (SHIPS), Type 1, environmental vibration. The detected output of the tube shall be displayed on an oscilloscope which shall be calibrated prior to vibration in the dc mode using slow sweep by means of a calibrated attenuator. The face of the display tube shall be marked with lines corresponding to the saturated power output level and to levels ± 0.5 dB therefrom. The attenuator then shall be returned to the saturated level. The displayed variations attributable to vibration shall be within the calibration lines.

22/ The shock test shall be conducted in accordance with MIL-STD-202, method 213, shock (specified pulse), using the sawtooth shock pulse, figure 213-2, with peak acceleration of 40 g, and duration of 10 ms. The tube thereafter shall be tested for conformance inspection, part 1, large-signal power output, and shall operate within the specified limits.

23/ Tubes selected for this test shall previously have passed the shock and vibration tests.

24/ The tube shall be operated intermittently according to the following schedule for the duration of the life test:

Heat voltage, Ef, only	4 hours
Test condition (1)	8 hours

Life hours will be based on filament hours.

25/ Pi(rf) is the incident power to the input r-f terminal of the tube. Unless otherwise specified, under test condition 1, Pi(rf) shall be that rf input power required to obtain saturation power output, Pso, at the specified frequency.

26/ Three tubes are required for these tests. Conformance inspection, part 1, shall be performed prior to and upon completion of qualification inspection.

27/ All tubes shall be tested in accordance with conformance inspection, part 1.

28/ Initial groups life test shall be conducted for 5,000 hours in the following sequence:

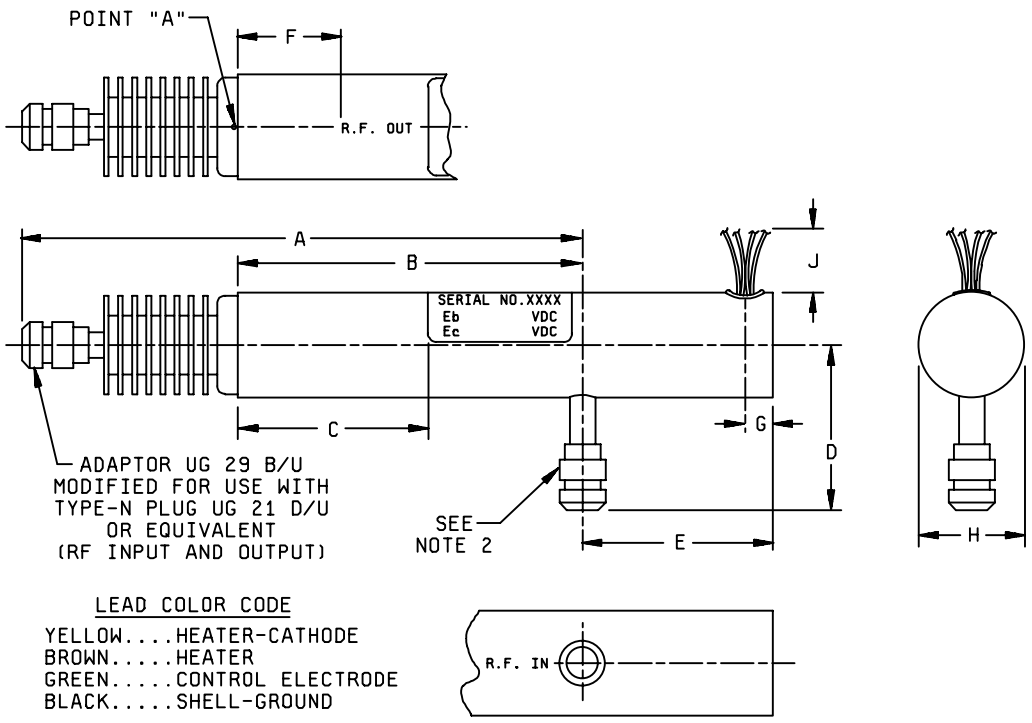
a. Operate in accordance with conformance inspection, part 3, for 336 ± 36 hours.

b. Test in accordance with conformance inspection, part 1, and conformance inspection, part 2, power output vibration.

c. Operate in accordance with conformance inspection, part 3, for 72 hours at F = 5.9 GHz and simultaneously operate in accordance with conformance inspection, part 2.

The “endurance test” of MIL-STD-167 (SHIPS) shall apply. The vibration frequency shall be the resonant value determined from conformance inspection, part 2, vibration test, or 33 Hz, with amplitude as shown in MIL-STD-167, table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:13Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	7.62	7.75	193.55	196.85
B	5.19	5.31	131.83	134.95
C	2.87	3.13	72.90	79.50
D	2.31	2.44	58.72	61.98
E	2.66	2.72	67.56	69.09
F	1.37	1.63	34.80	41.40
G	0.25	0.50	6.35	12.70
H	1.47	1.53	37.34	38.86
J	24.00	---	60.96	---

NOTES:
1. Point “A” = location where envelope temperature is measured.
2. Maximum permissible diameter of any portion of rf input = 0.740.

FIGURE 1. Outline drawing of electron tube type 8364 .

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:13Z
Check the source to verify that this is the current version before use.

Custodian:
Navy - EC

Review activities:
Navy - AS, CG, MC, OS

Preparing activity:
DLA - CC

(Project 5960-3503)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:13Z
Check the source to verify that this is the current version before use.